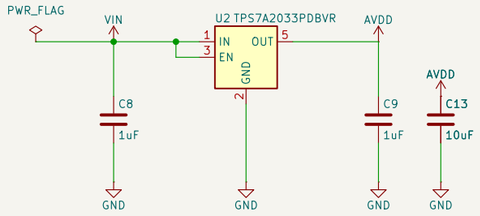
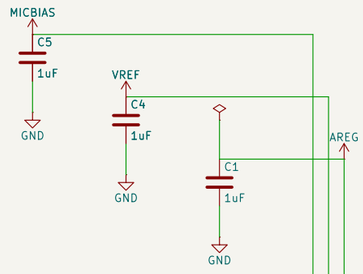
BirdNET microphone HAT

Analog microphone + onboard PDM microphone | TLV320ADC5140 | XIAO ESP32S3Plus

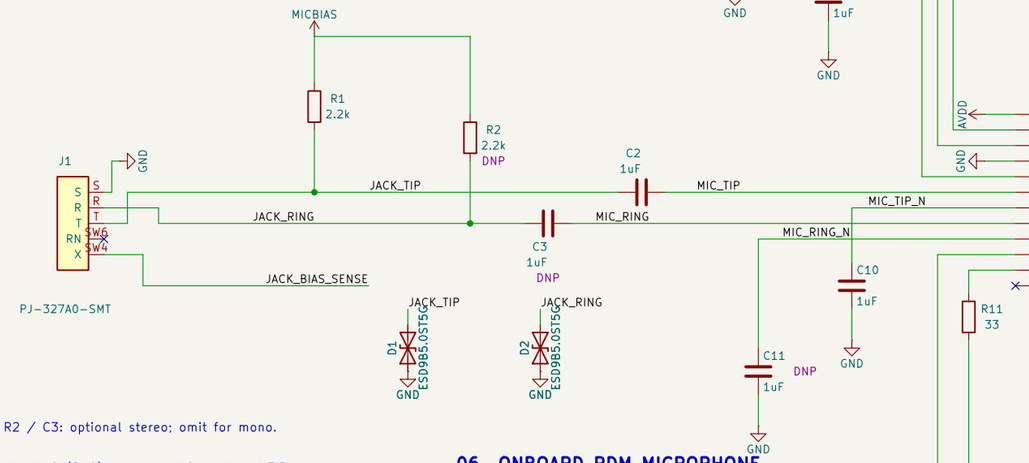
01 LOW-NOISE ANALOG SUPPLY



02 ADC REFERENCE / BIAS SUPPORT



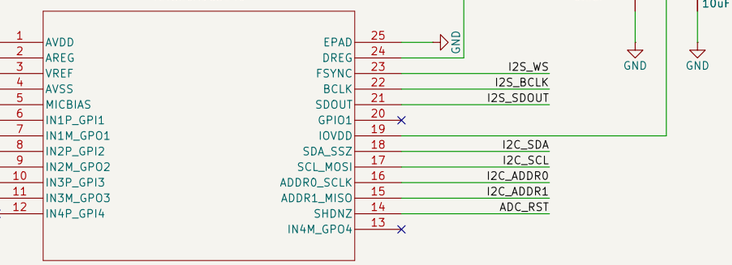
03 ANALOG JACK / AC COUPLING



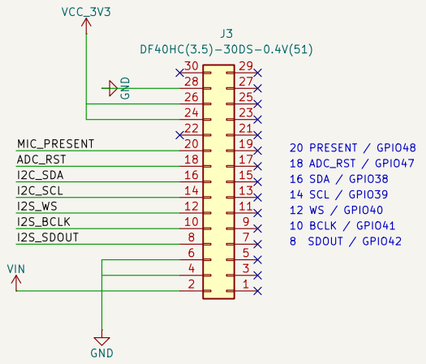
R2 / C3: optional stereo; omit for mono.

J1 pin 4 (SW4): assumed NC to biased TIP.
Verify with the actual jack; remove R6 to disable detection.

04 AUDIO ADC

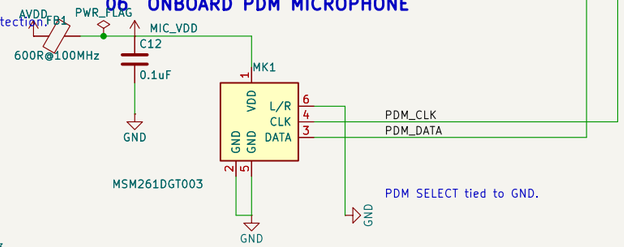


05 HOST INTERFACE

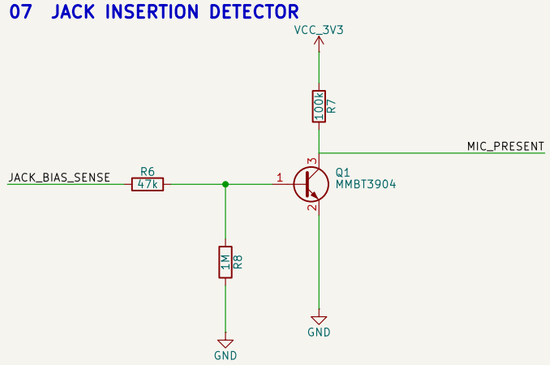


Pin 20: MIC_PRESENT / GPIO48 / active HIGH

06 ONBOARD PDM MICROPHONE

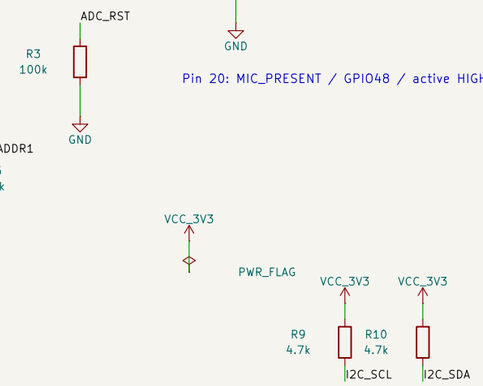


07 JACK INSERTION DETECTOR



MICBIAS enabled: no plug = Q1 on / MIC_PRESENT LOW.
Plug inserted: tip switch opens / MIC_PRESENT HIGH.
GPIO48: input, internal pulls disabled; debounce in firmware.

Enable MICBIAS and wait for settling before reading detection.
Remove R6 to isolate detection if contact mapping is wrong;
then ignore MIC_PRESENT in firmware (it remains HIGH).



R4/R5 hold the address straps low.

Crossing wires connect only where a junction dot is shown.
Named rails and detector interfaces join by their net labels.

Sheet: /
File: birdnet-mic-hat.kicad_sch

Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.7 | Id: 1/1